

K2 Physics Validation Report

Generated 2026-08-13 11:02 UTC

18 passed · 1 failed · 1 skipped

Each engine is benchmarked against an *independent* reference, in three tiers of increasing strength:

- 1. **Exact solutions** — the 6DOF integrator against closed-form ODE solutions, structures against textbook formulas, the cone solver against Taylor–Maccoll.
- 2. **Code-to-code** — flight against OpenRocket, aerodynamics against SU2, the airframe FEM against CalculiX.
- 3. **Published measurement** — 3 benchmarks compare K2 against wind-tunnel data from the open literature. Agreement with another program proves consistency; agreement with a measurement is the only tier that can show the physics is right. The table below lists these alongside the other published references (exact-solution tables, agreed benchmark target values), which are external but are not measurements.

Comparisons against published data

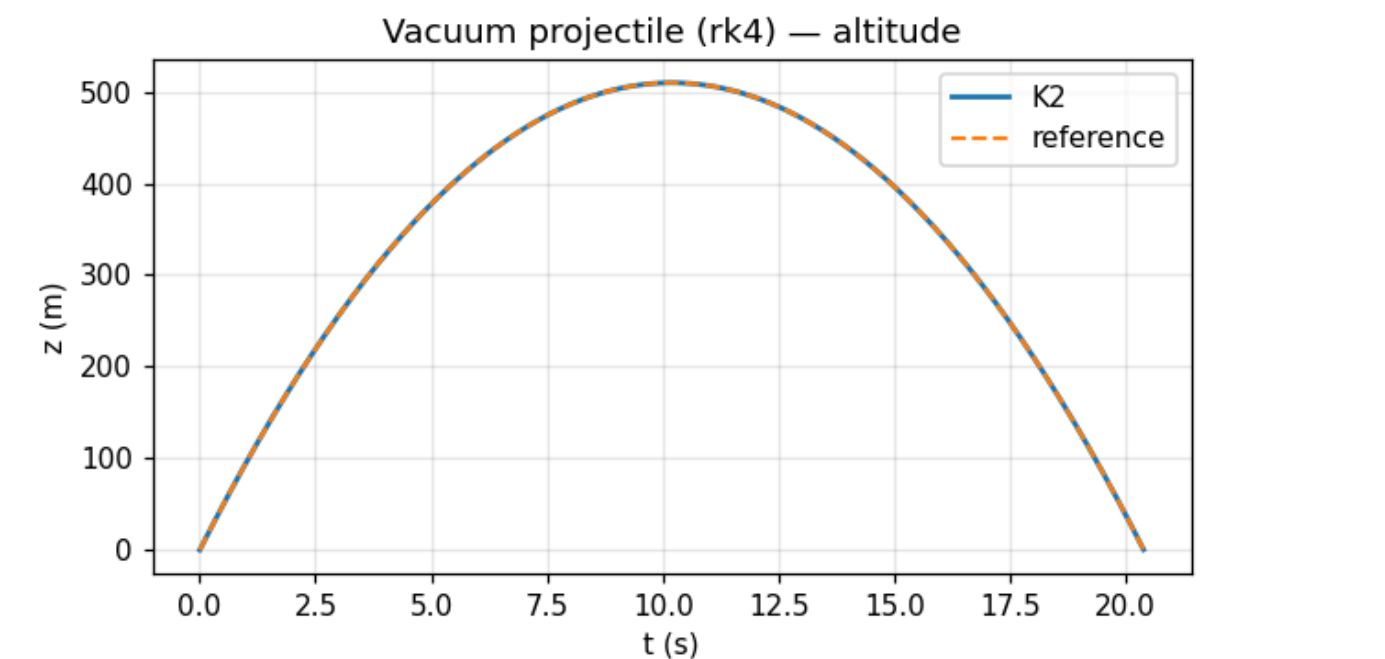
Benchmark	Source	Kind	Status
Taylor–Maccoll cone vs NACA-1135	NACA Report 1135 cone tables	published exact-solution tables	PASS
Barrowman lift slope vs AGARD-B experiment	AEDC-TR-70-100 wind-tunnel data (AGARD-B, Tunnel 4T)	wind-tunnel measurement	PASS
SU2 lift slope vs AGARD-B experiment	AEDC-TR-70-100 wind-tunnel data (AGARD-B, Tunnel 4T)	wind-tunnel measurement	PASS
ONERA M6 wing surface Cp vs experiment	Schmitt & Charpin, AGARD AR-138 (1979), Test 2308	wind-tunnel measurement	FAIL
NAFEMS LE1 — elliptic membrane (CalculiX)	NAFEMS standard benchmark target value	agreed benchmark target value	PASS
NAFEMS LE10 — thick plate under pressure (CalculiX)	NAFEMS standard benchmark target value	agreed benchmark target value	PASS

Flight Simulation (6DOF) ↔ Integrator exact solutions / OpenRocket

PASS — Vacuum projectile (rk4)

Reference: Exact kinematics $z=v_0t-\frac{1}{2}g\cdot t^2$ | Credibility: Validated

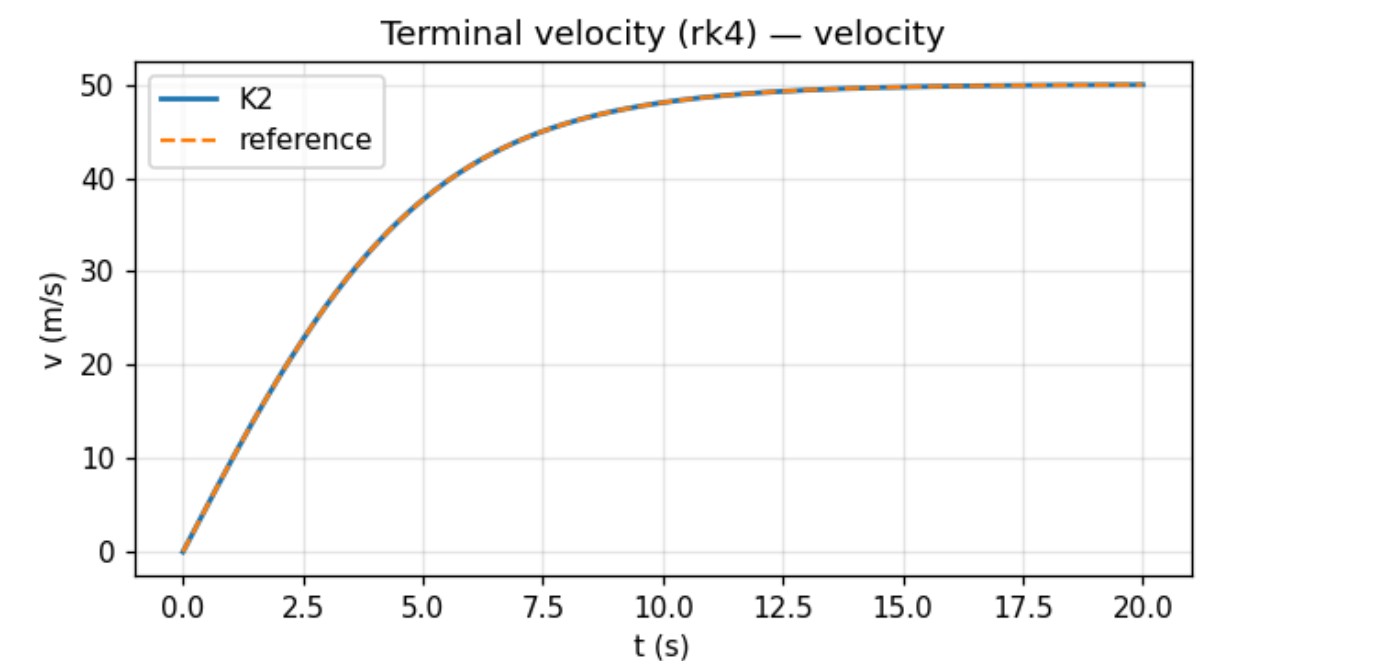
Quantity	K2	Reference	Source	Rel. err	Tol	Status
Apogee height	509.9	509.9	$v_0^2/2g$	0.00%	0%	✓
Position RMSE vs parabola	1.492e-11	0	exact	0.00%	/ 0.05	✓



PASS — Terminal velocity (rk4)

Reference: Exact $v(t) = v_t \tanh(g \cdot t / v_t)$, $v_t = \sqrt{g/k}$ | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Terminal velocity	49.96	50	$\sqrt{g/k}$	0.08%	0%	✓
Velocity RMSE vs tanh	5.968e-12	0	exact	0.00%	/ 0.02	✓



PASS — Oscillator energy (rk4)

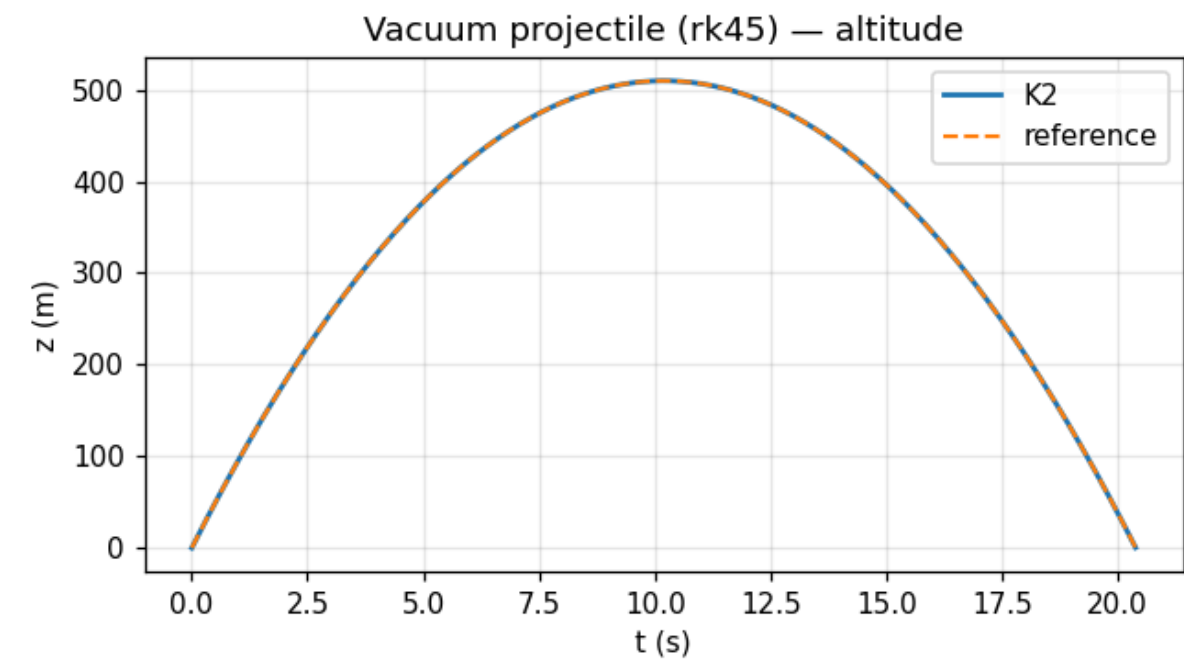
Reference: Conserved $E = \frac{1}{2}v^2 + \frac{1}{2}\omega^2 x^2$ over 50 periods | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Energy drift (50 periods)	2	2	$E(0)$	0.00%	1%	✓

PASS — Vacuum projectile (rk45)

Reference: Exact kinematics $z=v_0t-\frac{1}{2}g\cdot t^2$ | Credibility: Validated

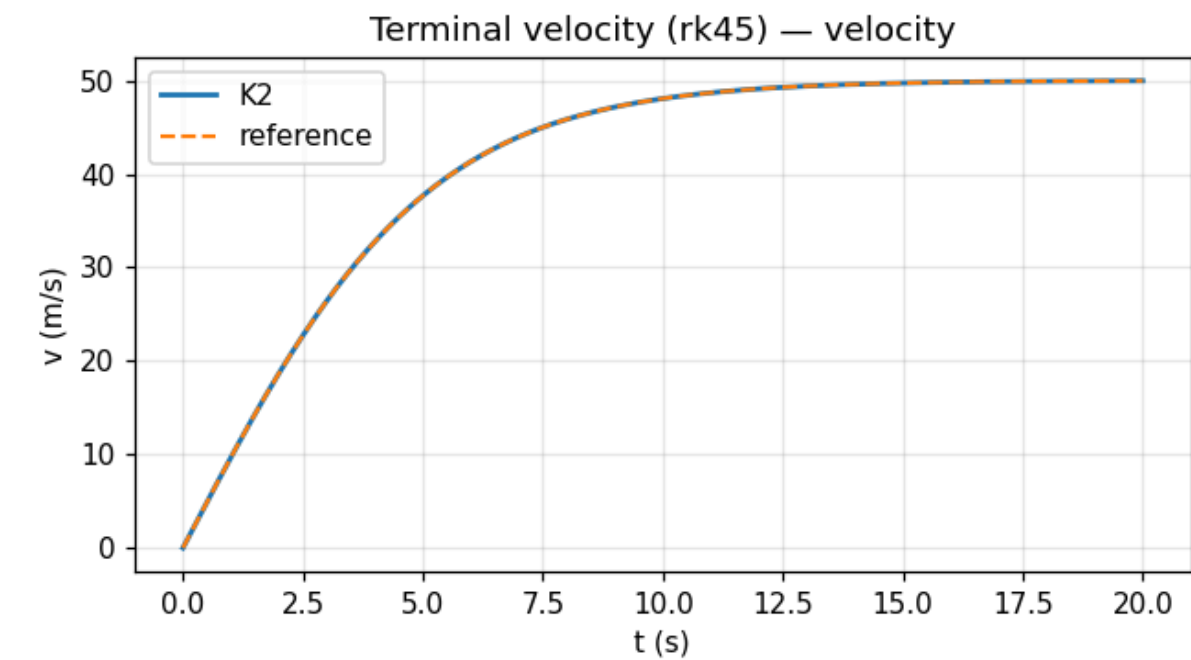
Quantity	K2	Reference	Source	Rel. err	Tol	Status
Apogee height	509.9	509.9	$v_0^2/2g$	0.00%	0%	✓
Position RMSE vs parabola	1.493e-11	0	exact	0.00%	/ 0.05	✓



PASS — Terminal velocity (rk45)

Reference: Exact $v(t)=v_t\cdot\tanh(g\cdot t/v_t)$, $v_t=\sqrt{g/k}$ | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Terminal velocity	49.96	50	$\sqrt{g/k}$	0.08%	0%	✓
Velocity RMSE vs tanh	1.423e-13	0	exact	0.00%	/ 0.02	✓



PASS — Oscillator energy (rk45)

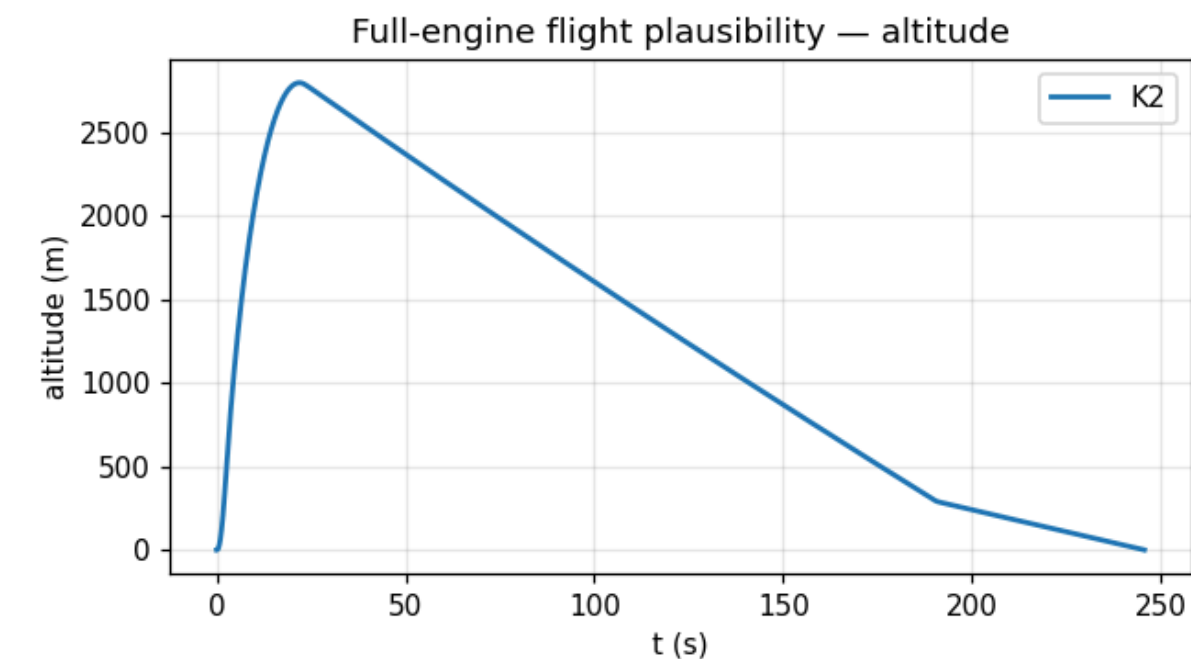
Reference: Conserved $E = \frac{1}{2}v^2 + \frac{1}{2}\omega^2 x^2$ over 50 periods | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Energy drift (50 periods)	2	2	E(0)	0.00%	1%	✓

PASS — Full-engine flight plausibility

Reference: Physical sanity bounds (canonical L1090W rocket) | Credibility: Estimated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Apogee in [500, 6000] m	1	1	bound	0.00%	/ 0.5	✓
Vehicle landed	1	1	bound	0.00%	/ 0.5	✓
Max-Q before apogee	1	1	bound	0.00%	/ 0.5	✓
Subsonic-to-low-supersonic max Mach	1	1	bound	0.00%	/ 0.5	✓



Sim vs OpenRocket (*skipped*)

Reference: OpenRocket flight CSV (canonical rocket)

Reason: set OPENROCKET_CSV to an OpenRocket flight-data CSV of the canonical rocket (build it per validation/cases/rocket_canonical.py, motor L1090W, and export flight data)

Aerodynamics ↔ Wind-tunnel measurement / Taylor–Maccoll exact / SU2

PASS — Taylor–Maccoll cone vs NACA-1135

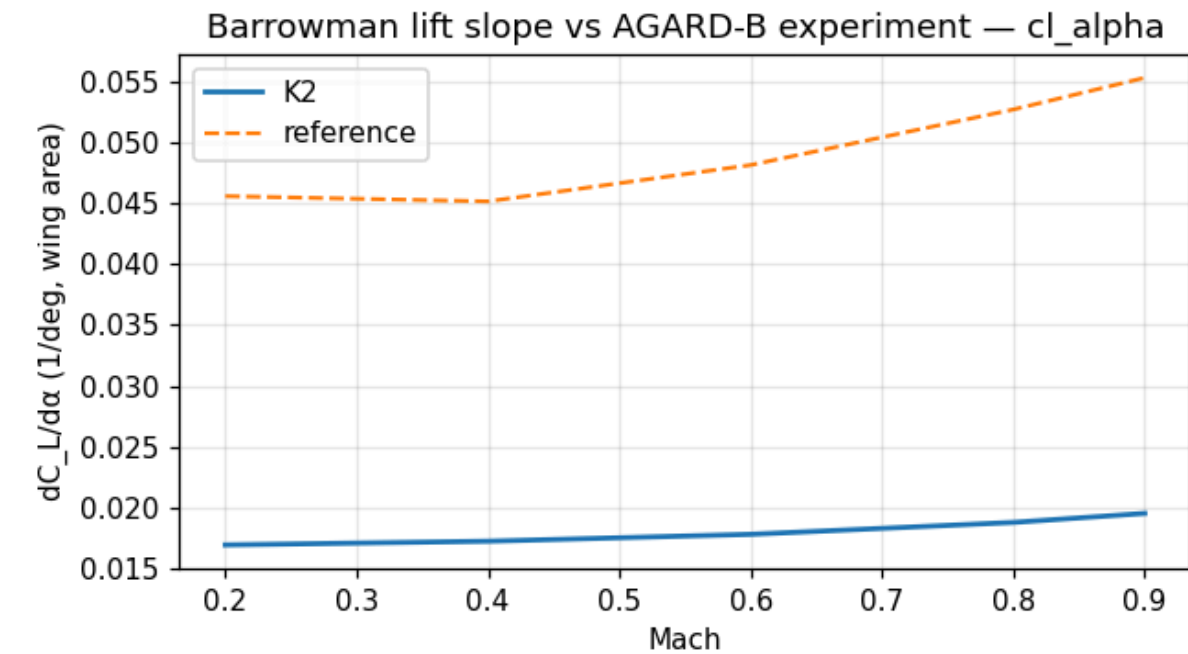
Reference: NACA Report 1135 cone tables | *Credibility:* Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Shock angle ($M=2.0$, $\theta_c=10.0^\circ$)	31.21	31.2	NACA-1135	0.02%	1%	✓
Surface C_p ($M=2.0$, $\theta_c=10.0^\circ$)	0.1045	0.105	NACA-1135	0.50%	5%	✓
Shock angle ($M=3.0$, $\theta_c=10.0^\circ$)	21.71	21.8	NACA-1135	0.39%	1%	✓
Surface C_p ($M=3.0$, $\theta_c=10.0^\circ$)	0.08748	0.088	NACA-1135	0.59%	5%	✓

PASS — Barrowman lift slope vs AGARD-B experiment

Reference: AEDC-TR-70-100 wind-tunnel data (AGARD-B, Tunnel 4T) | *Credibility:* Estimated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
$C_{L\alpha}(M=0.4) / C_{L\alpha}(M=0.2)$	1.018	0.9906	AEDC-TR-70-100	2.76%	8%	✓
$C_{L\alpha}(M=0.6) / C_{L\alpha}(M=0.2)$	1.052	1.056	AEDC-TR-70-100	0.43%	8%	✓
$C_{L\alpha}(M=0.8) / C_{L\alpha}(M=0.2)$	1.11	1.156	AEDC-TR-70-100	4.05%	8%	✓
$C_{L\alpha}(M=0.9) / C_{L\alpha}(M=0.2)$	1.153	1.214	AEDC-TR-70-100	4.96%	8%	✓
$C_{L\alpha}$ at $M=0.2$ (absolute, diagnostic)	0.0169	0.0456	AEDC-TR-70-100	62.93%	300%	✓
$C_{L\alpha}$ at $M=0.6$ (absolute, diagnostic)	0.01778	0.04816	AEDC-TR-70-100	63.09%	300%	✓
$C_{L\alpha}$ at $M=0.9$ (absolute, diagnostic)	0.0195	0.05534	AEDC-TR-70-100	64.77%	300%	✓



PASS — SU2 cone vs Taylor–Maccoll

Reference: Taylor–Maccoll exact (M=2, 10° cone) | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Cone surface Cp (M=2, 10°)	0.1044	0.1045	Taylor–Maccoll	0.06%	10%	✓

PASS — Barrowman aero vs SU2

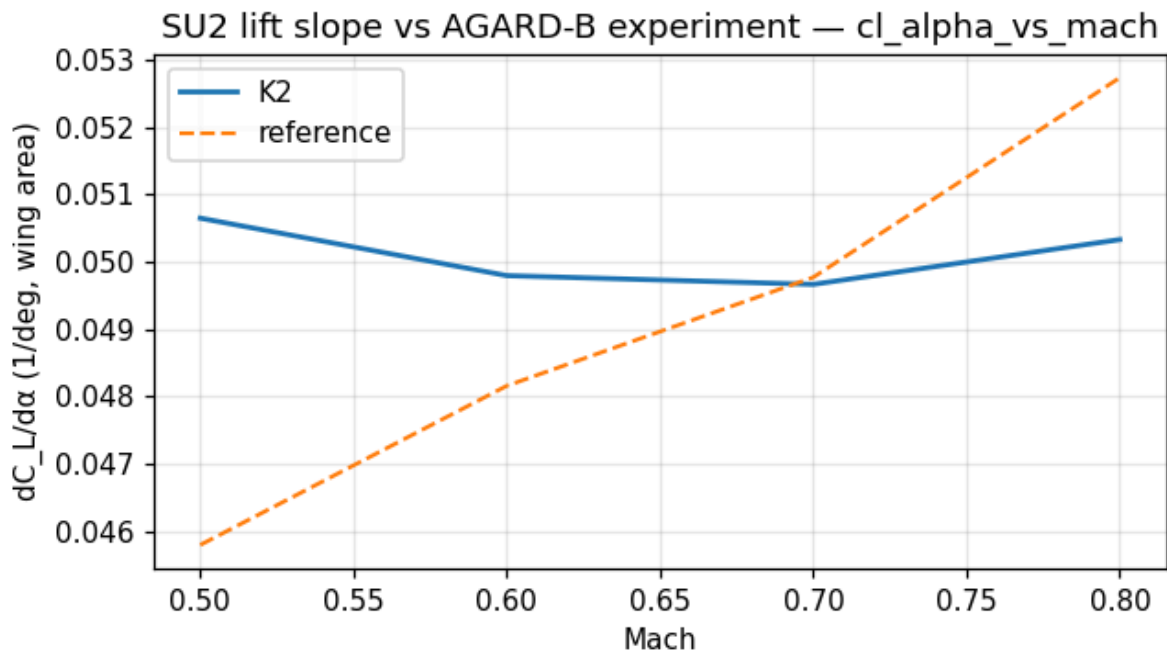
Reference: SU2 RANS (canonical rocket) | Credibility: Estimated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
CFD reference area	0.008171	0.008171	$\pi \cdot (d_body/2)^2$	0.00%	2%	✓
Drag positive	1	1	sign	0.00%	/ 0.5	✓
Lift positive at +AoA	1	1	sign	0.00%	/ 0.5	✓
Drag coefficient Cd (diagnostic)	0.433	0.7889	SU2	45.11%	100%	✓
Normal-force Cn vs SU2 Cl (diagnostic)	0.9122	3.11	SU2 (Cl)	70.67%	100%	✓

PASS — SU2 lift slope vs AGARD-B experiment

Reference: AEDC-TR-70-100 wind-tunnel data (AGARD-B, Tunnel 4T) | Credibility: Validated

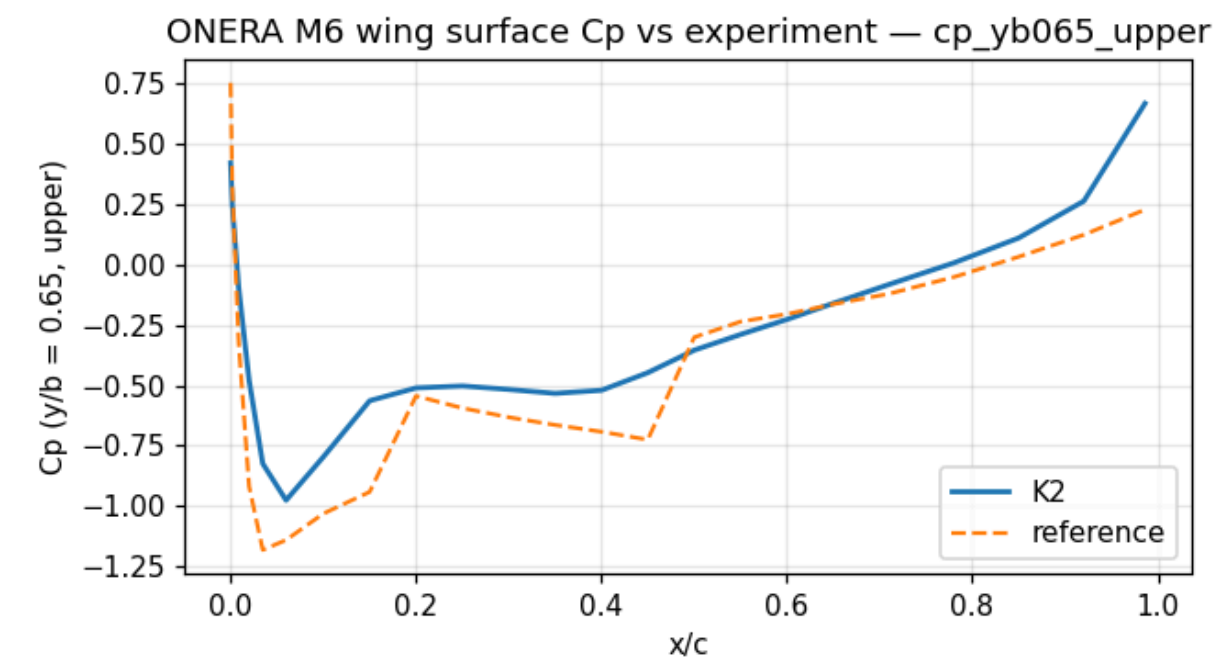
Quantity	K2	Reference	Source	Rel. err	Tol	Status
$dC_L/d\alpha$ at M=0.5	0.05065	0.0458	AEDC-TR-70-100	10.58%	15%	✓
C_L at M=0.5, $\alpha=6^\circ$	0.2997	0.2761	AEDC-TR-70-100	8.56%	20%	✓
$dC_L/d\alpha$ at M=0.6	0.04979	0.04816	AEDC-TR-70-100	3.39%	15%	✓
C_L at M=0.6, $\alpha=6^\circ$	0.2947	0.2934	AEDC-TR-70-100	0.43%	20%	✓
$dC_L/d\alpha$ at M=0.7	0.04967	0.04977	AEDC-TR-70-100	0.21%	15%	✓
C_L at M=0.7, $\alpha=6^\circ$	0.2939	0.3038	AEDC-TR-70-100	3.26%	20%	✓
$dC_L/d\alpha$ at M=0.8	0.05033	0.05273	AEDC-TR-70-100	4.55%	15%	✓
C_L at M=0.8, $\alpha=6^\circ$	0.2981	0.3217	AEDC-TR-70-100	7.34%	20%	✓



FAIL — ONERA M6 wing surface C_p vs experiment

Reference: Schmitt & Charpin, AGARD AR-138 (1979), Test 2308 | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Cp RMSE improves with refinement, y/b=0.2 (upper)	0.1487	0.2012	AGARD AR-138	26.10%	≤ ref (15%)	✓
Cp RMSE improves with refinement, y/b=0.2 (lower)	0.08755	0.1206	AGARD AR-138	27.39%	≤ ref (15%)	✓
Cp RMSE improves with refinement, y/b=0.65 (upper)	0.2176	0.2681	AGARD AR-138	18.85%	≤ ref (15%)	✓
Cp RMSE improves with refinement, y/b=0.65 (lower)	0.06179	0.08375	AGARD AR-138	26.21%	≤ ref (15%)	✓
Cp RMSE improves with refinement, y/b=0.9 (upper)	0.2791	0.3246	AGARD AR-138	14.01%	≤ ref (15%)	✓
Cp RMSE improves with refinement, y/b=0.9 (lower)	0.06968	0.09676	AGARD AR-138	27.99%	≤ ref (15%)	✓
Cp RMSE improves with refinement, y/b=0.99 (upper)	0.3248	0.3826	AGARD AR-138	15.12%	≤ ref (15%)	✓
Cp RMSE improves with refinement, y/b=0.99 (lower)	0.1321	0.188	AGARD AR-138	29.75%	≤ ref (15%)	✓
Sectional normal force c_n , y/b=0.2	0.1741	0.2359	AGARD AR-138	26.19%	15%	✗
Sectional normal force c_n , y/b=0.65	0.1617	0.2841	AGARD AR-138	43.08%	15%	✗
Sectional normal force c_n , y/b=0.9	0.1179	0.2131	AGARD AR-138	44.68%	15%	✗
Sectional normal force c_n , y/b=0.99	0.1279	0.2106	AGARD AR-138	39.28%	15%	✗



Structures ↔ NAFEMS benchmarks / textbook closed form / CalculiX

PASS — Closed-form structural formulas

Reference: Textbook thin-wall / Euler relations | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Hoop stress σ_h	4e+07	4e+07	p·r/t	0.00%	2%	✓
Euler buckling P_{cr}	2.752e+05	2.752e+05	$\pi^2 EI/L^2$	0.00%	5%	✓
von Mises (uniaxial)	1.5e+08	1.5e+08	σ	0.00%	0%	✓

PASS — Uniaxial bar (CalculiX)

Reference: Exact $\delta = FL/EA$, $\sigma = F/A$ | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Tip elongation δ	1.797e-05	1.814e-05	FL/EA	0.95%	3%	✓
Axial stress σ (mean)	2.5e+06	2.5e+06	F/A	0.00%	2%	✓

PASS — Cantilever bending (CalculiX)

Reference: Euler–Bernoulli $\delta = FL^3/3EI$ | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Tip deflection δ	0.002242	0.002268	$FL^3/3EI$	1.11%	5%	✓

PASS — 1st bending mode: closed-form vs CalculiX

Reference: CalculiX modal (cantilever) | Credibility: Estimated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
f1 (1st lateral bending)	25.81	25.78	CalculiX mode 1	0.11%	25%	✓

PASS — NAFEMS LE1 — elliptic membrane (CalculiX)

Reference: NAFEMS standard benchmark target value | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
σ_{yy} at point D	9.294e+07	9.27e+07	NAFEMS	0.26%	5%	✓
σ_{yy} at D (integration points, diagnostic)	7.949e+07	9.27e+07	NAFEMS	14.25%	40%	✓

PASS — NAFEMS LE10 — thick plate under pressure (CalculiX)

Reference: NAFEMS standard benchmark target value | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
σ_{yy} at point D	-5.596e+06	-5.38e+06	NAFEMS	4.01%	10%	✓
σ_{yy} at D (integration points, diagnostic)	-4.282e+06	-5.38e+06	NAFEMS	20.40%	40%	✓